Burbury, Ernst Zermelo, and Willard Gibbs—Ludwig Boltzmann came to appreciate that the full story of time's arrow is more surprising. Boltzmann realized that although entropy had illuminated important aspects of the puzzle, it had *not* answered the question of why the past and the future seem so different. Instead, entropy had redefined the question in an important way, one that leads to an unexpected conclusion.

Entropy: Past and Future

Earlier, we introduced the dilemma of past versus future by comparing our everyday observations with properties of Newton's laws of classical physics. We emphasized that we continually experience an obvious directionality to the way things unfold in time but the laws themselves treat what we call forward and backward in time on an exactly equal footing. As there is no arrow within the laws of physics that assigns a direction to time, no pointer that declares, "Use these laws in this temporal orientation but not in the reverse," we were led to ask: If the laws underlying experience treat both temporal orientations symmetrically, why are the experiences themselves so temporally lopsided, always happening in one direction but not the other? Where does the observed and experienced directionality of time come from?

In the last section we seemed to have made progress, through the second law of thermodynamics, which apparently singles out the future as the direction in which entropy increases. But on further thought it's not that simple. Notice that in our discussion of entropy and the second law, we did not modify the laws of classical physics in any way. Instead, all we did was use the laws in a "big picture" statistical framework: we ignored fine details (the precise order of *War and Peace*'s unbound pages, the precise locations and velocities of an egg's constituents, the precise locations and velocities of a bottle of Coke's CO_2 molecules) and instead focused our attention on gross, overall features (pages ordered vs. unordered, egg splattered vs. not splattered, gas molecules spread out vs. not spread out). We found that when physical systems are sufficiently complicated (books with many pages, fragile objects that can splatter into many fragments, gas with many molecules), there is a huge